

LECTURE 10 KEY POINTS

- When two variables are **correlated** or they show some form **association**, we cannot infer **causation**.
Not necessarily one is the **cause** of the other.
(see cases 2, 3, 5, 7)
- When two variables are **associated**, it might be because both **depend** on a third variable which has a **causal relationship** with both. In this case we call the third variable **CONFOUNDING VARIABLE**.
(see cases 2, 3, 5, 7)
- The **SIMPSON'S PARADOX** is when two variables show a certain association without taking into account a specific **confounding variable**, but when we take into account such **confounder** the association is completely reversed (see case 2)
- Always use both summary measures and plots when studying one variable or two variables jointly.
Try and support your intuitions with statistical evidence.